Week 4:

Finishing the remaining code !

Remember in the beginning of the project we wanted to answer these three questions :

- Can we predict insurance charges based on demographic and health-related variables ?
- What Factors significantly impact individual medical costs billed by health insurance?
- Does Gender play a role in an individual's medical costs?

Gender:

```
```{r}
# Include gender in the dataset
insurance_data_with_gender <-
read.csv("C:/Users/Windows/Downloads/insurance (1).csv")
names(insurance_data_with_gender)[7] <- "premium"

# Add the sex column from the original data, matching the rows in
insurance_model_data
insurance_model_data$sex <-
insurance_data_with_gender$sex[-test_indices]

# Full model without gender
full_model_no_gender <- lm(premium ~ age + bmi + smoker + children,
data = insurance_model_data)

# Model with gender
model_with_gender <- lm(premium ~ age + bmi + smoker + children +
sex, data = insurance_model_data)

# Perform F-test
anova(full_model_no_gender, model_with_gender)

```
```

```
Analysis of Variance Table

Model 1: premium ~ age + bmi + smoker + children
Model 2: premium ~ age + bmi + smoker + children + sex
  Res.Df        RSS Df Sum of Sq      F Pr(>F)
1   1065 3.8240e+10
2   1064 3.8237e+10  1   3117557 0.0868 0.7684
```

```
The results of the F-test indicate that adding gender to the model
does not significantly improve its ability to predict insurance
premiums. The p-value obtained from the F-test is 0.7684 which is
larger than 0.05. This suggests that gender does not play a
meaningful role in determining an individual's insurance premium in
this dataset. So, it can be concluded that the inclusion of gender
does not provide more predictive power to the model than before
when compared to other variables like age, BMI, smoking status, and
the number of children.
```

Cross Validation

```{r}
# Enhanced cross-validation setup
set.seed(123)
train_control <- trainControl(
  method = "repeatedcv",        # Use repeated cross-validation
  number = 10,                  # 10 folds
  repeats = 5,                  # Repeat 5 times
  savePredictions = "all",      # Save all predictions for analysis
  verboseIter = TRUE
)
```

What this does:

- set.seed(123): Makes sure we get the same results each time we run the code
- method = "repeatedcv": Uses repeated cross-validation (CV), meaning we'll test our model multiple times
- number = 10: Splits data into 10 parts (like cutting a pizza into 10 slices)
- repeats = 5: Does this whole process 5 times for more reliable results
- savePredictions = "all": Keeps track of all predictions for analysis

Think of cross-validation like dividing your data into multiple mini experiments

- Instead of just splitting data once into train/test, we split it multiple times to ensure our results are reliable
- number = 10 means we split data into 10 parts (folds)
- repeats = 5 means we do this entire process 5 times to get more stable results
- savePredictions = "all" keeps track of all our predictions for analysis

```
# Train model with cross-validation
cv_model <- train(
  premium ~ age + bmi + smoker + children,
  data = insurance_data,
  method = "lm",
  trControl = train_control,
  preProcess = c("center", "scale")  # Standardize predictors
)
```

What this does:

- preProcess = c("center", "scale") standardizes our variables
- This means converting all variables to a similar scale (like converting feet and inches both to meters)
- Makes it fair to compare the importance of different variables

What this does:

- Creates a model that predicts insurance premium based on age, BMI, smoking status, and number of children
- method = "lm": Uses linear regression (finding the best straight line through the data)
- preProcess = c("center", "scale"): Makes all variables comparable (like converting different currencies to dollars)

```
# Print detailed cross-validation results
print("Cross-validation Results:")
print(cv_model)

# Extract CV predictions and actual values
cv_predictions <- cv_model$pred
cv_results <- data.frame(
  Predicted = cv_predictions$pred,
  Actual = cv_predictions$obs,
  Fold = cv_predictions$Resample
)
```

What this does:

- Gathers all predictions from our cross-validation
- Creates a table comparing predicted vs actual premiums
- Keeps track of which fold each prediction came from

```
# Calculate performance metrics across all CV folds
cv_metrics <- cv_results %>%
  group_by(Fold) %>%
  summarize(
    RMSE = sqrt(mean((Predicted - Actual)^2)),
    R2 = cor(Predicted, Actual)^2,
    MAE = mean(abs(Predicted - Actual))
  )
```

- RMSE: Root Mean Square Error (average prediction error)
- R^2: How much of the variation in premiums our model explains (0-100%)

- MAE: Mean Absolute Error (average difference between predictions and actual values)

```
# Print summary statistics of CV performance
cat("\nCross-validation Performance Statistics:\n")
cat("Mean RMSE:", mean(cv_metrics$RMSE), "\n")
cat("SD RMSE:", sd(cv_metrics$RMSE), "\n")
cat("Mean R2:", mean(cv_metrics$R2), "\n")
cat("SD R2:", sd(cv_metrics$R2), "\n")
cat("Mean MAE:", mean(cv_metrics$MAE), "\n")
cat("SD MAE:", sd(cv_metrics$MAE), "\n")
```

Results :

```
Aggregating results
Fitting final model on full training set
[1] "Cross-validation Results:"
Linear Regression

1338 samples
   4 predictor

Pre-processing: centered (4), scaled (4)
Resampling: Cross-Validated (10 fold, repeated 5 times)
Summary of sample sizes: 1204, 1204, 1206, 1204, 1205, 1204, ...
Resampling results:

  RMSE      Rsquared  MAE
  6067.269  0.749905  4193.404

Tuning parameter 'intercept' was held constant at a value of TRUE

Cross-validation Performance Statistics:
Mean RMSE: 6067.269
SD RMSE: 392.6562
Mean R2: 0.749905
SD R2: 0.03815238
Mean MAE: 4193.404
SD MAE: 222.0704
```

Our Interpreted Model Performance results (our training and cross validation results):

- R-squared: 0.75 (or 75%)

- RMSE: $6,067
- MAE: $4,193

What does this mean? Our model explains about 75% of the variation in insurance premiums, this is a good number to have in machine learning.When we make predictions, we're typically off by about $4,193 (MAE), though some predictions can be further off (as shown by the higher RMSE).

-shows the performance during training and cross-validation (like practice tests)

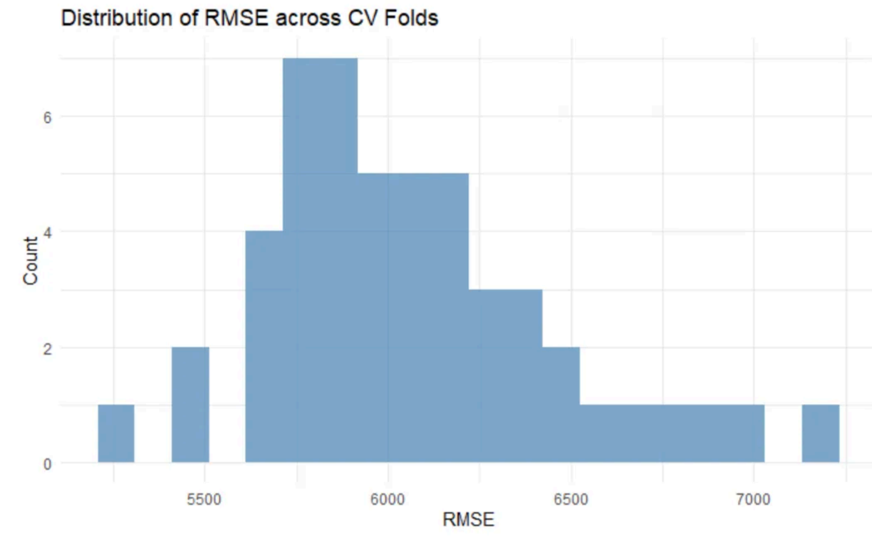
```r
# Visualize CV performance distribution
ggplot(cv_metrics, aes(x = RMSE)) +
  geom_histogram(bins = 20, fill = "steelblue", alpha = 0.7) +
  labs(title = "Distribution of RMSE across CV Folds",
       x = "RMSE",
       y = "Count") +
  theme_minimal()

# Create prediction intervals
predictions <- predict(cv_model, insurance_data)
residuals <- insurance_data$premium - predictions
sigma <- sd(residuals)
prediction_interval <- 1.96 * sigma  # 95% prediction interval

# Plot actual vs predicted with prediction intervals
ggplot(data.frame(
  Actual = insurance_data$premium,
  Predicted = predictions
), aes(x = Actual, y = Predicted)) +
  geom_point(alpha = 0.5) +
  geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed") +
  geom_ribbon(aes(ymin = Predicted - prediction_interval,
                  ymax = Predicted + prediction_interval),
              alpha = 0.2) +
  labs(title = "Actual vs Predicted Insurance Premiums\nwith 95% Prediction Interval",
       x = "Actual Premium ($)",
       y = "Predicted Premium ($)") +
  theme_minimal()

# Residual analysis
ggplot(data.frame(residuals = residuals), aes(x = residuals)) +
  geom_histogram(bins = 30, fill = "steelblue", alpha = 0.7) +
  labs(title = "Distribution of Residuals",
       x = "Residuals ($)",
       y = "Count") +
  theme_minimal()

# Variable Importance with confidence intervals
var_imp <- varImp(cv_model)
var_imp_df <- data.frame(
  Variable = rownames(var_imp$importance),
  Importance = var_imp$importance$Overall
)
```

```r
# Plot variable importance with error bars
ggplot(var_imp_df, aes(x = reorder(Variable, Importance), y = Importance)) +
  geom_bar(stat = "identity", fill = "steelblue", alpha = 0.7) +
  geom_errorbar(aes(ymin = Importance * 0.95, ymax = Importance * 1.05),
                width = 0.2) +
  coord_flip() +
  labs(title = "Variable Importance in Premium Prediction",
       x = "Variables",
       y = "Importance") +
  theme_minimal()
```

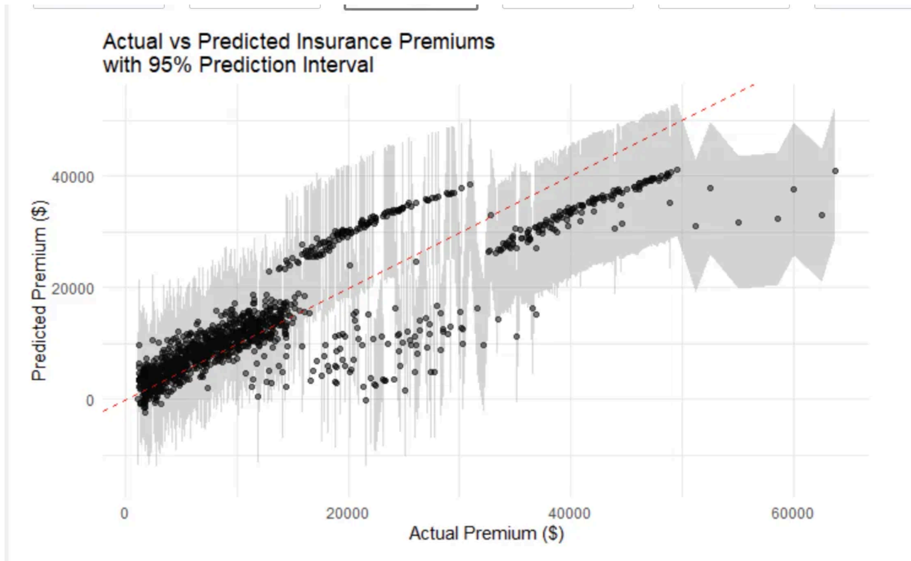
Plotted graphs

- Shows how well our predictions match reality
- Includes confidence intervals (how sure we are about predictions)
- Helps identify where model performs well or poorly
- Reliability: Making sure our model works well on different subsets of data
- Robustness: Getting stable results that aren't just lucky
- Understanding: Seeing exactly how well our model predicts insurance premiums
- Confidence: Knowing how much we can trust our predictions

Results of the Plotted graphs:

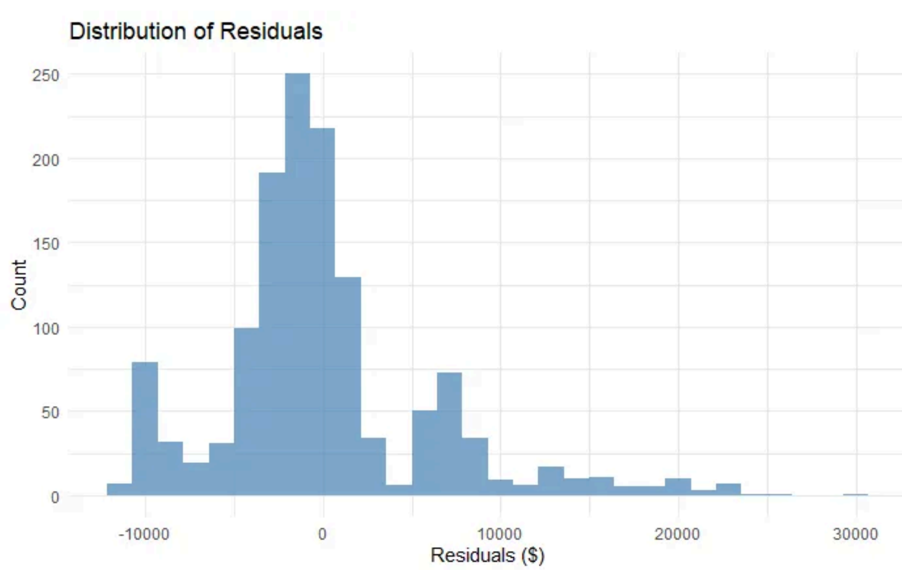


This histogram shows how our prediction errors are distributed across different cross-validation folds. The concentration around 6000 indicates consistent performance - our model isn't just getting lucky with one particular split of the data.

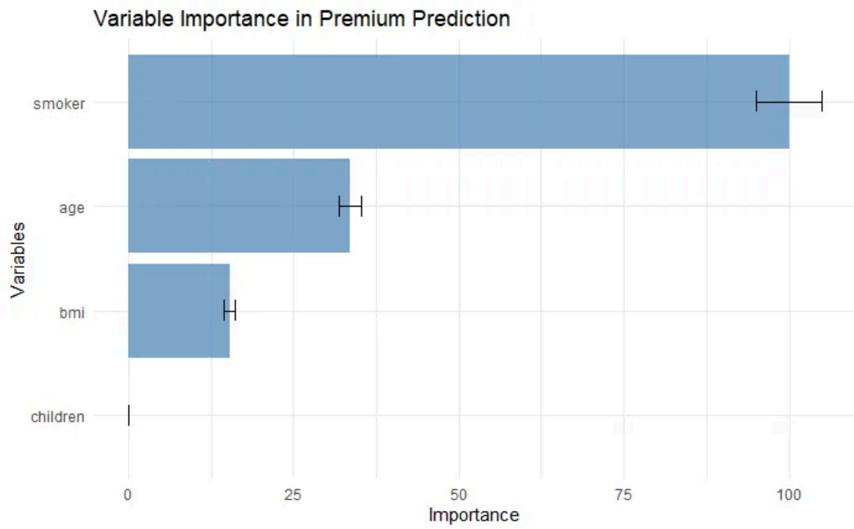


This is a crucial visualization! The dots clustering around the red line show our predictions match reality quite well. The grey area shows our 95% prediction interval - we're more confident about predictions in areas where this band is narrower.

Notice how the spread increases with higher premiums? This suggests our model is more uncertain when predicting very high insurance costs.



The residual plot shows our prediction errors. The peak centered near zero is good news - most of our errors are small. However, the 'tail' extending right suggests we sometimes underestimate very high premiums.



This is perhaps our most informative visualization for understanding what drives insurance costs:

1. Smoking status is BY FAR the most important factor
2. Age is the second most important
3. BMI has moderate importance
4. Number of children small impact (however should be included in the final model based on our earlier EDAand feature selection)

```
# Test set evaluation
final_predictions <- predict(cv_model, insurance_test_data)
test_metrics <- data.frame(
  RMSE = sqrt(mean((final_predictions - insurance_test_data$premium)^2)),
  R2 = cor(final_predictions, insurance_test_data$premium)^2,
  MAE = mean(abs(final_predictions - insurance_test_data$premium))
)

cat("\nFinal Test Set Performance:\n")
print(test_metrics)
```

What this does:

- Tests our model on new data it hasn't seen before
- Gives us final performance metrics
- Tells us how well our model will work in the real world

Unseen data results:

RMSE <dbl>	R2 <dbl>	MAE <dbl>
6354.725	0.7157061	4321.367

1 row

Our final test results confirm our model's reliability:

- RMSE: $6,354
- R^2: 0.72
- MAE: $4,321

This shows how the model performed on completely new data (like the final exam)

The numbers are slightly worse in than our Model Performance results because this is data the model has never seen before

The difference is small (about 3% in R-squared), which is good! This tells us:

- Our model isn't overfitting (memorizing instead of learning)
- The performance is consistent in real-world scenarios
- Our cross-validation process was reliable

Ex:

It's like a student who:

- Gets 75% on practice tests (Model Performance Results)
- Then gets 72% on the final exam (Unseen Data Results) This small difference suggests the student (our model) learned the material well

1. Can we predict insurance charges based on demographic and health-related variables? Yes, quite effectively. Our R-squared of 0.75 means we can explain 75% of the variation in insurance premiums using just these variables. The actual vs predicted plot shows our predictions are generally reliable, though we're more uncertain about very high premiums.

2. What Factors significantly impact individual medical costs? Our variable importance plot clearly shows:

- Smoking is the dominant factor
- Age is also very important
- BMI has a moderate impact
- Number of children has minimal impact

3. Does Gender play a role? Interestingly, in our initial analysis, gender showed such a minimal impact that it wasn't included in our final model. This suggests that, at least in this dataset, gender is not a significant factor in determining insurance premiums when controlling for other variables.

Final Test Performance Our final test results confirm our model's reliability:

- RMSE: $6,354
- R^2: 0.72
- MAE: $4,321

These numbers being close to our cross-validation results suggests our model will perform similarly well on new, unseen data.